

EXPERIMENT NO. 2

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TITLE

NumPy Basics – Student Marks Analysis

OBJECTIVE

To familiarize with NumPy array equations by storing the marks of 3 students in 5 subjects and finding the total, average, highest score and subject-wise average.

AIM

To store the marks of 3 students in 5 subjects in a NumPy array and compute the total, average, highest score and subject-wise average.

THEORY

Marks matrix (3 students $\times$ 5 subjects):

$$M = \begin{bmatrix} m_{11} & \cdots & m_{15} \\ m_{21} & \cdots & m_{25} \\ m_{31} & \cdots & m_{35} \end{bmatrix}$$

Total of all marks:

$$T = \sum_{i=1}^3 \sum_{j=1}^5 m_{ij} \quad (\text{np.sum})$$

Overall average:

$$\bar{A} = \frac{T}{15} \quad (\text{np.mean})$$

Highest score and its position:

$$H = \max(M) \quad (\text{np.max, np.argmax})$$

Student-wise total (row sum, axis=1):

$$T_i = \sum_{j=1}^5 m_{ij}$$

Subject-wise average (column mean, axis=0):

$$\bar{S}_j = \frac{1}{3} \sum_{i=1}^3 m_{ij}$$

PROCEDURE

1. Store the marks of 3 students in 5 subjects in a 3×5 NumPy array.
2. Find the total (np.sum) and the average (np.mean) of all marks.
3. Find the highest score (np.max) and its student/subject (np.argmax).
4. Find the student-wise total (sum, axis=1) and subject-wise average (mean, axis=0).
5. Display all results and verify them.

DATASET DESCRIPTION

Dataset Name: Student marks (synthetic, entered in program)

Source: 3 students $\times$ 5 subjects, marks out of 100

Number of Samples: 15 marks Number of Features: 5 subjects

Target / Class: Not applicable (descriptive statistics)

Description: S1=[78, 85, 92, 88, 76]; S2=[65, 72, 80, 69, 74]; S3=[90, 88, 95, 91, 89].

PROGRAM

```
import numpy as np

marks = np.array([
    [78, 85, 92, 88, 76],
    [65, 72, 80, 69, 74],
    [90, 88, 95, 91, 89],
])

print("Marks (3 students x 5 subjects):")
print(marks)

total = marks.sum()
average = marks.mean()
highest = marks.max()
pos = np.unravel_index(marks.argmax(), marks.shape)
print("Total =", total)
print("Average =", round(float(average), 2))
print("Highest score =", highest, "by student", pos[0] + 1,
      "in subject", pos[1] + 1)

stud_total = marks.sum(axis=1)
print("Student-wise total =", stud_total)
sub_avg = marks.mean(axis=0)
print("Subject-wise average =", np.round(sub_avg, 2))
```

OUTPUT

Marks (3 students x 5 subjects):

```
[[78 85 92 88 76]
 [65 72 80 69 74]
 [90 88 95 91 89]]
```

Total = 1232

Average = 82.13

Highest score = 95 by student 3 in subject 3

Student-wise total = [419 360 453]

Subject-wise average = [77.67 81.67 89. 82.67 79.67]

RESULT

The above program was implemented and executed successfully using Python, and the corresponding output was obtained and verified. The results of the **NumPy array statistics** were analyzed and found to be satisfactory, thereby achieving the desired objective of the experiment.

Faculty Signature: _____

Date: _____